Practical 2

Electricity and Electronics EBN 111

24 – 28 May 2021

Practical Information			
Maximum marks:	40	Full marks:	40
Duration of practical:	180 min	Open/Closed book:	Open
Total number of pages (this page included)		13	

Lecturers: Dr D. le Roux, Dr S. Smith, Dr X. Ye

Assistant Lecturers: M. Maritz, J. Masela, J. Thiselton, M. Trivedi, D. Wepener, L. Kokot

Important

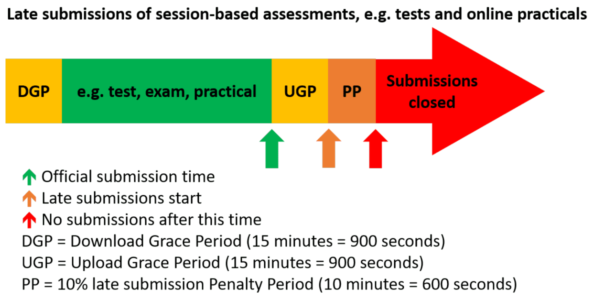
1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write irrational numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By uploading your practical, you declare the following:**

   The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment.

Online assessment submission rules and time-line

The following rules and time-line apply for all practicals for EBN111/122 as illustrated by the figure below:

1. The Download Grace Period (DGP) is a 15 minute period before the official start of the practical for students to download all documents related to the practical and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the practical.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the practical.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2021.practicals@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers during this period without incurring any penalties. No email submissions will be accepted during the UGP.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheets during this period, the student incurs a 10% penalty to their final practical mark.
8. No late submissions is accepted after the PP.



Please Note:

TinkerCAD Circuits may round some numbers that are entered into the different instruments. As long as you make sure you enter the value exactly into the blue parameter box in TinkerCAD Circuits, the simulation will be correct, even if the instrument display shows a rounded value. This applies to all questions in the practical.

Questions 1 to 4 [8]

Overview:

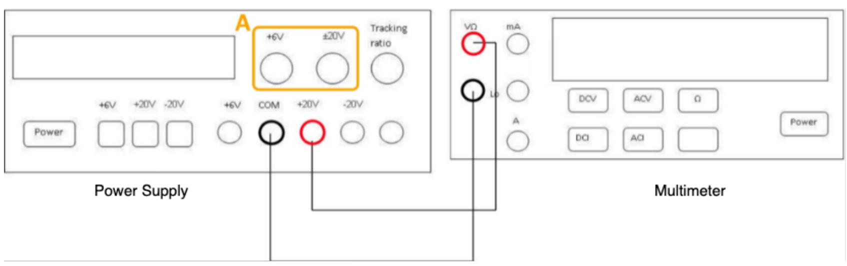
This question introduces the power supply and the multimeter. The goal here is to understand that a measured value and a theoretical value are not always exactly the same.

Instructions:

Please note: You will not implement steps 1 to 3 on TinkerCAD Circuits, but must understand the settings and functionality of the real laboratory equipment and be able to apply this to carry out steps 4 to 9.

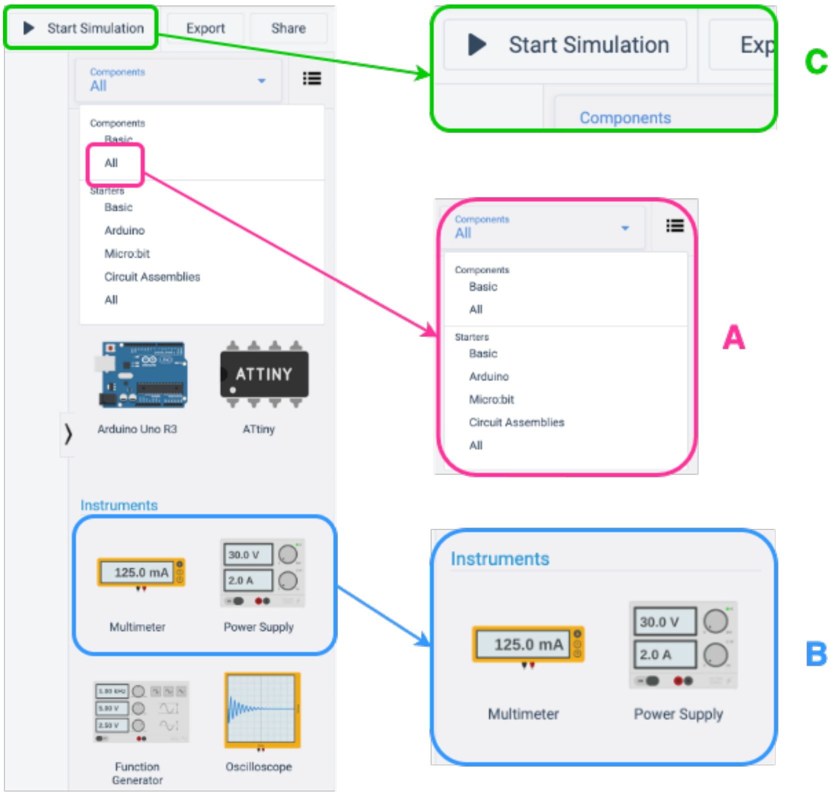
1. When doing experiments in a real lab with real equipment, before you switch on the power supply, make sure all the voltage knobs are turned down to the minimum. This will prevent damage to the equipment and your circuit when you switch on the power. The voltage is decreased by rotating the voltage knob counter clockwise and increased by rotating the knob clockwise (marked A in Figure 1).
2. To measure the voltage supplied by the power supply, connect the power supply directly to the multimeter as shown in Figure 1. Do so by connecting the positive red terminals to each other and the negative black terminals to each other. Remember that you are measuring DC voltage, so make sure the multimeter is set to measure this by pressing the DCV button on the multimeter.
3. After setting up the equipment, you can switch on the power supply and read the voltage on the multimeter.

Figure 1:



4. To replicate the voltage measurement shown in Figure 1 without real equipment, you will use the DC power supply and multimeter in TinkerCAD Circuits. You can find them by selecting "All" under the "Components" drop-down list (marked A in Figure 2). Then scroll down and look for "Power Supply" and "Multimeter" under the "Instruments" label (marked B in Figure 2). You can also use the search bar to search for components. Click and drag the required equipment to the workspace.

Figure 2:



5. Connect the power supply and the multimeter using the real-world laboratory setup shown in Figure 1 as a reference.
6. Click on the multimeter and select "Voltage" under the mode drop-down list.
7. Click on the power supply and enter the voltage with $V_{ps} = [V_{ps}]$.
8. Click on "Start Simulation" to run the simulation (marked C in Figure 2).
9. Simulate and measure V_{mm} for each $[V_{ps}]$ value given. (Note: the measured voltage value will not necessarily correspond EXACTLY with the power supply output voltage value!).

What is the voltage V_{mm} measured using the multimeter for the $[V_{ps}]$ values given below?

01	V_{mm1} if $V_{ps} = [V_{ps1}]$ [2]
02	V_{mm2} if $V_{ps} = [V_{ps2}]$ [2]
03	V_{mm3} if $V_{ps} = [V_{ps3}]$ [2]
04	V_{mm4} if $V_{ps} = [V_{ps4}]$ [2]

Questions 5 to 6 [4]

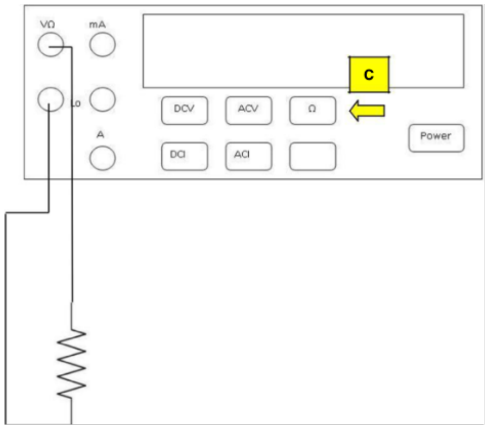
Overview:

This question involves measuring resistance combinations using the multimeter. (Please note: For the labelling of resistors: A $1k2\ \Omega$ resistor is $1200\ \Omega$. Be careful not to mix up $12\ \Omega$, $1k2\ \Omega$ and $12k\ \Omega$).

Instructions:

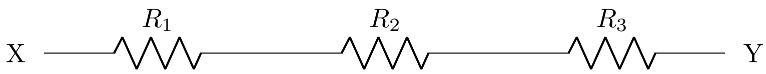
In the laboratory, resistor values can be measured as shown in Figure 3 below. Button C as indicated in Figure 3 is pressed to measure the resistance. This set-up can be replicated in TinkerCAD Circuits.

Figure 3:



1. The values for the resistors in Figures 4 and 5 are $R_1 = [R_1]$, $R_2 = [R_2]$ and $R_3 = [R_3]$.
2. Using a virtual breadboard in TinkerCAD Circuits, build the resistor combinations as indicated in Figures 4 and 5.
3. Measure the equivalent resistance R_{eq} of the combinations between points X and Y in TinkerCAD Circuits using the multimeter. Ensure the multimeter is set to "Resistance" mode.

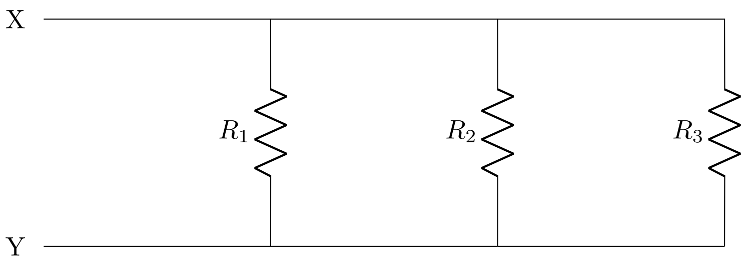
Figure 4:



What is the equivalent resistance R_{eq}?

05	R_{eq1} for Figure 4 [2]

Figure 5:



What is the equivalent resistance R_{eq}?

06	R_{eq2} for Figure 5 [2]

Questions 7 to 8 [4]

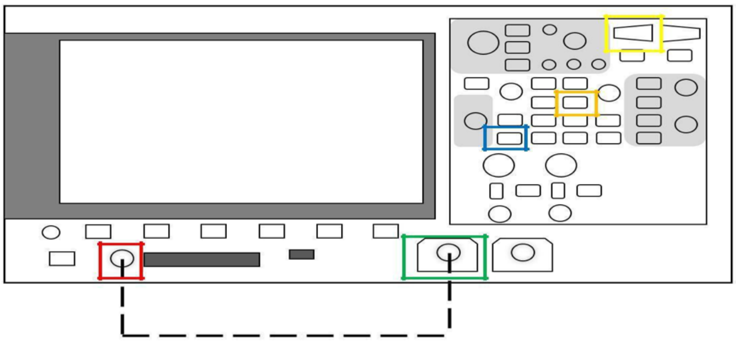
Overview:

In this question you will monitor a signal generated by a function generator using the oscilloscope in TinkerCAD Circuits.

Instructions:

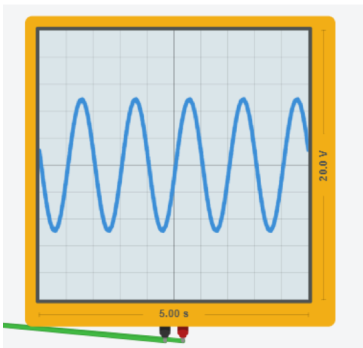
In the laboratory, the function generator and oscilloscope form part of the same piece of equipment as shown in Figure 6.

Figure 6:



1. To replicate Figure 6 without real equipment, you will use two separate components in TinkerCAD Circuits, namely the "Function generator" and the "Oscilloscope". You can find them by selecting "All" under the "Components" drop-down list (marked A in Figure 2). Then scroll down and look for "Function generator" and "Oscilloscope" under the "Instruments" label. You can also use the search bar to search for components. Click and drag the required equipment to the workspace.
2. Figure 7 shows an example of an oscilloscope result obtained using TinkerCAD Circuits. The values shown on the x and y axes are the values across the entire length of each axis. For the example in Figure 7, this means that each block (or division) is 0.5 s along the time axis, while each division is 2 V along the vertical voltage axis. Parameters of the generated waveform can thus be measured using both horizontal and vertical readings.

Figure 7:



3. Connect the output ports of the function generator to the input ports of the oscilloscope using wires. Do so by connecting the positive red terminals to each other and the negative black terminals to each other.
4. Click on the function generator and set the waveform to be a sine wave using the Function drop-down menu.

5. Set the function generator frequency to [f], amplitude to [A] and offset to 0 V.

   Please note: The Amplitude setting on the function generator is the peak-to-peak amplitude, which is the difference between the maximum and minimum voltage of the input signal.

6. Click on the oscilloscope and set the Time per Division ([t/div]).

7. Run the simulation and wait for a clear sine wave to appear on the oscilloscope.

8. Measure the amplitude and period.

What is the amplitude A_{out} (zero to peak) of the sine wave?	
07	A_{out} [2]
What is the period T_{out} (one full cycle) of the sine wave?	
08	T_{out} [2]

Questions 9 to 10 [4]

Overview:

In this question you will test the principle of voltage division. Figure 8 illustrates the concept. You will solve the circuit theoretically and then measure the value of V_2 by building the circuit using TinkerCAD Circuits, with a set-up similar to Figure 9.

Instructions:

Figure 8:

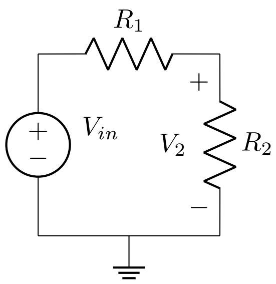
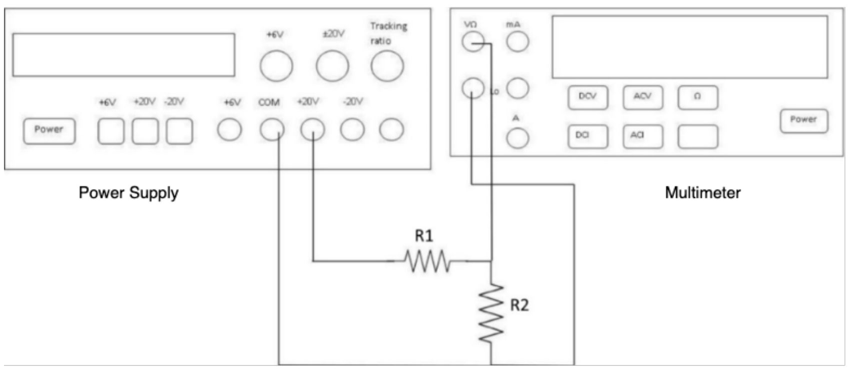


Figure 9:



1. Calculate V_{2t}, the theoretical value of V_2, for Figure 8 by using Ohm's law and voltage division whereby $V_{in} = [V_{int}]$, $R_1 = [R_{1t}]$, and $R_2 = [R_{2t}]$.

What is the theoretical value of the voltage V_2?

09	V_{2t} [2]

2. The practical values $V_{in} = [V_{inp}]$, $R_1 = [R_{1p}]$, and $R_2 = [R_{2p}]$ are slightly different to the theoretical values to account for component tolerances.
3. Build the circuit on the breadboard in TinkerCAD Circuits and set your power supply voltage, $V_{in} = [V_{inp}]$, and resistor values, $R_1 = [R_{1p}]$, and $R_2 = [R_{2p}]$, to the practical values. Use the multimeter in TinkerCAD Circuits to measure the voltage. Refer to Figure 9 for how to connect the resistors and instruments to measure the voltage V_2.
4. Run the simulation and measure the voltage V_{2p}, the practical value of V_2, as obtained for Figure 8 and built using TinkerCAD Circuits.

What is the measured value of the voltage V_2?

10	V_{2p} [2]

Questions 11 to 14 [8]

Overview:

In this question you will test the concept of current division. You will solve the circuit of Figure 10 theoretically and then build the circuit using TinkerCAD Circuits, with a set-up similar to Figure 11.

Instructions:

Figure 10:

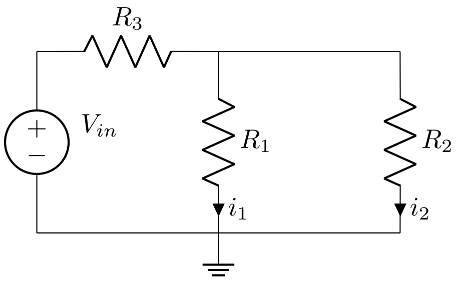
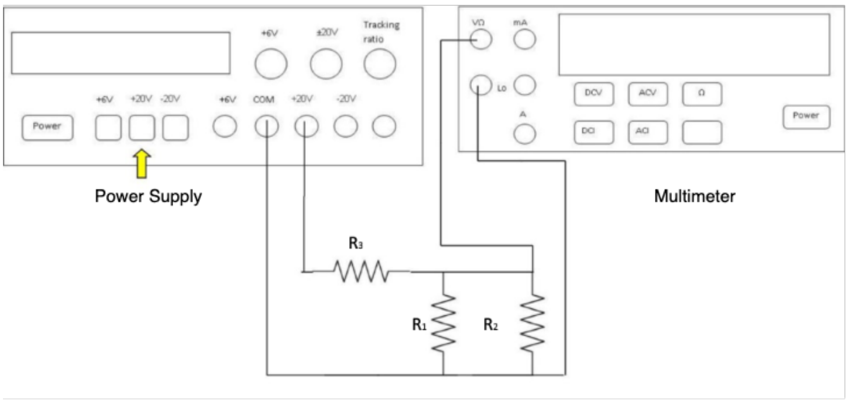


Figure 11:



1. Calculate i_{1t} and i_{2t}, the theoretical values of i_1 and i_2, for Figure 10 using Ohm's law and current division whereby $V_{in} = [V_{int}]$, $R_1 = [R_{1t}]$, $R_2 = [R_{2t}]$ and $R_3 = [R_{3t}]$.

What is the theoretical value of the current i_{1t}?

11	i_{1t} [2]

What is the theoretical value of the current i_{2t}?

12	i_{2t} [2]

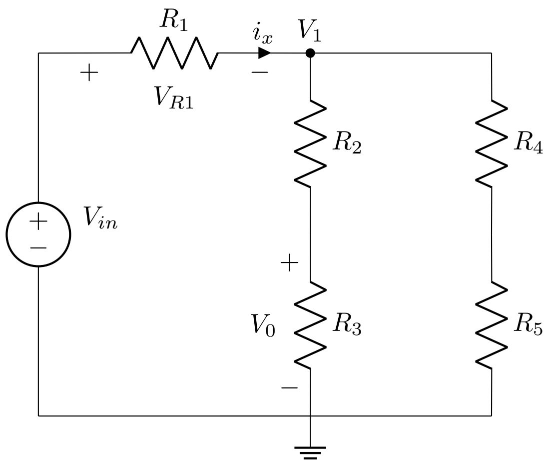
2. The practical values $V_{in} = [V_{inp}]$, $R_1 = [R_{1p}]$, $R_2 = [R_{2p}]$ and $R_3 = [R_{3p}]$ are slightly different to the theoretical values to account for component tolerances.

3. Build the circuit on the breadboard in TinkerCAD Circuits and set your power supply voltage, $V_{in} = [V_{inp}]$, and resistor values $R_1 = [R_{1p}]$, $R_2 = [R_{2p}]$ and $R_3 = [R_{3p}]$, to the practical values. Measure the voltage across resistors R_1 and R_2. Refer to Figure 11 for how to connect the resistors and instruments to measure the voltage.

4. Run the simulation and calculate the measured currents i_{1p} and i_{2p}, the practical values of i_1 and i_2, **using Ohm's Law (first measure the voltage and resistance and then apply Ohm's law)** as obtained for Figure 10 and built using TinkerCAD Circuits.

What is the measured value of the current i_{1p}?	
13	i_{1p} [2]
What is the measured value of the current i_{2p}?	
14	i_{2p} [2]

Questions 15 to 17 [6]	
Overview: In this question you will further develop the circuit of the previous question and investigate various aspects such as the current through and voltage across the components.	
Instructions: **Figure 12:**	



1. The values in Figure 12 are $V_{in} = [V_{in}]$, $R_1 = [R_1]$, $R_2 = [R_2]$, $R_3 = [R_3]$, $R_4 = [R_4]$ and $R_5 = [R_5]$.
2. Build the circuit shown in Figure 12 on the breadboard in TinkerCAD Circuits and measure the voltage across R_1. Use this value to determine i_x with **Ohm's Law** and measure the voltages V_1 and V_0.

What is the value of the current i_x?	
15	i_x [2]
What is the value of the voltage V_1?	
16	V_1 [2]
What is the value of the voltage V_0?	
17	V_0 [2]

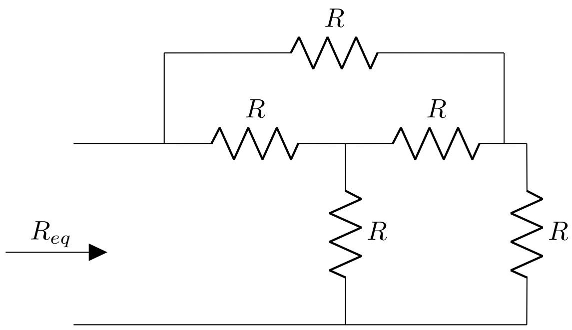
Question 18 [2]

Overview:

In this question you will investigate equivalent resistive networks in slightly greater detail than in questions 5 and 6.

Instructions:

Figure 13:



1. The value for the resistance in Figure 13 is $R = [R]$.
2. Build the circuit shown in Figure 13 on the breadboard in TinkerCAD Circuits and measure the equivalent resistance R_{eq} using the multimeter.

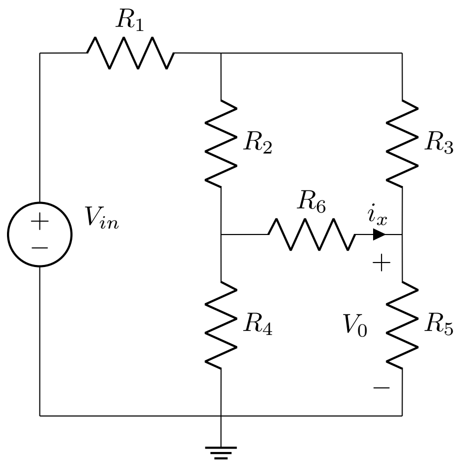
What is the equivalent resistance R_{eq}?

18	R_{eq} [2]

Questions 19 to 20 [4]
Overview: In this question you will combine Ohm's law with measurements from TinkerCAD to determine the current within the circuit.

Instructions:

Figure 14:



1. The values corresponding to Figure 14 are $V_{in} = [V_{in}]$, $R_1 = [R_1]$, $R_2 = [R_2]$, $R_3 = [R_3]$, $R_4 = [R_4]$, $R_5 = [R_5]$ and $R_6 = [R_6]$.
2. Build the circuit shown in Figure 14 on the breadboard in TinkerCAD Circuits. Measure the values of i_x and V_0. Remember that to measure i_x, you can first measure the voltage over R_6 and then use Ohm's law to calculate the current given the measured value of R_6.

What is the value of the current i_x?	
19	i_x [2]
What is the value of the voltage V_0?	
20	V_0 [2]

Grand Total : [40]